

Extended Essay – Physics

**The Effect of Magnetic Field Strength on the Mechanical
Power Output of a DC Motor**

Research Question:

How does varying the magnetic field strength (B/T) of a DC motor affect its
mechanical power output (P/W)?

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Table of contents

1-Introduction 4

2-Background Information 5

3-Experimental Variables 12

4-Experimental Design..... 14

4.1-Apparatus 14

4.2-Method 15

4.3-Explanation for Apparatus..... 19

4.4-Safety Information 20

5-Raw Data Collection 20

6-Processed Data 21

6.1 - Experimental Data..... 21

6.2 – Theoretical Data 22

7 - Analysis 23

7.1 – Explaining the Trends..... 24

7.2 - Comparison of Experimental Data and Theoretical Models 28

8 – Evaluation and Conclusion 29

9 - Works Cited..... 32

10 – Appendices..... 34

10.1 - Appendix A1: Base Theoretical Simulink Model of DC Motor 34

10.2 - Appendix A2: Thermal Theoretical Simulink Model of DC Motor	34
10.3 - Appendix A3: Thermal Subsystem of Thermal Theoretical Simulink Model of DC Motor	35
10.4 - Appendix A4: Base Theoretical DC Motor Parameters	35
10.5 - Appendix A5: Thermal Theoretical DC Motor Parameters	36
10.6 - Appendix A6: Sample Calculations for Magnetic Field Strength Under Saturated Conditions	38
10.7 - Appendix A7: Rotational Speed Data from Theoretical Models	38
10.8 – Appendix A8: Conversion of Absolute to Percentage Uncertainties	39

1-Introduction

Within physics, the DC motor serves as a bridge between electromagnetism and mechanics. It is a practical demonstration of how a magnetic torque acts on a current-carrying conductor and how electric energy is converted to mechanical energy. Electric motors play an important role in contemporary society (Young et al., 2020, p. 902); therefore, investigating their properties is crucial to optimize performance and to develop a sustainable society through implementation in green technologies such as renewable energy and electric vehicles.

This extended essay attempts to answer the question: **“How does varying the magnetic field strength of a DC motor affect its mechanical power?”** The objective of this essay is to analyze the trends and shapes of the graphs of the data and making comparisons with the data from two theoretical DC motor simulations. MATLAB Simulink was used to construct the two DC motor simulations – one that served as a basic model of the DC motor’s behaviour, which would also serve as the saturation model, and another that incorporated thermal losses, all of which constructed with experimental parameters. This combination of practical experimentation with computational and theoretical frameworks aids in developing a comprehensive understanding of DC motor dynamics.

2-Background Information

In a motor a magnetic torque acts on current-carrying conductor, and electric energy is converted to mechanical energy (Young et al., 2020, p. 902). There exist two main types of motors: AC and DC motors. AC motors are electric motors that are driven by alternating current, whilst DC motors are electric motors that convert direct current into mechanical energy. This paper focuses primarily on DC motors.

DC motors find many applications across a wide range of industries and fields due to their ease of use, simplicity, controllability, and versatility. These motors can be found in various automotive applications, industrial conveyor belt systems, robotic arms, CNC machines, and various actuating and power systems, as well as in medical applications or in everyday devices like home appliances, consumer electronics and so forth (Kuczmann, 2024, p. 1).

Principle of Electromagnetic Induction

Recall that the total magnetic flux Φ_B through a finite surface area A is given by the equation:

$$\Phi_B = BA \cos \phi \quad (\text{eq. 1})$$

Where,

- B = magnetic field strength (T)
- A = area element (m^2)
- ϕ = angle between the vector perpendicular to the surface and the magnetic field lines

Faraday's law of electromagnetic induction explains how changing magnetic flux induces an electromotive force, as shown in experiments with DC motors. Young et al. (2020, p. 956) states:

$$\varepsilon = -\frac{d\Phi_B}{dt} \quad (\text{eq. 2})$$

Where,

- ε = induced electromotive force (emf) (V)
- Φ_B = magnetic flux through the loop (Wb)

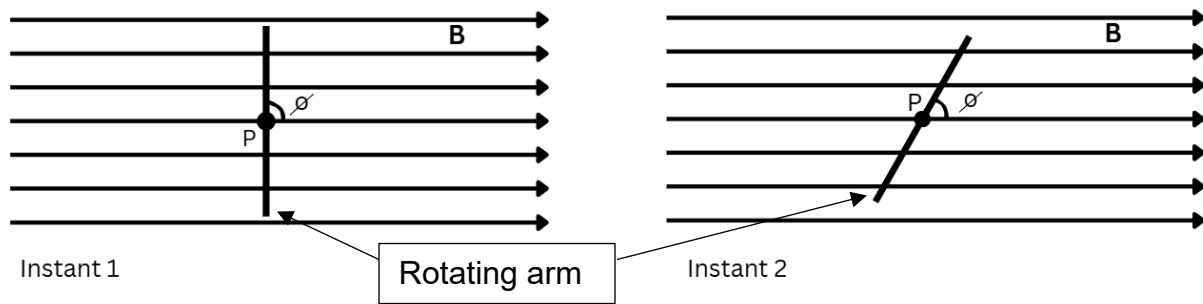


Figure 1.1: diagram illustrating change in magnetic flux between two instants of time.

In Figure 1.1 above, the magnetic flux passing through the area of the rotating arm pivoting at P decreases from instant 1 to instant 2, since the angle ϕ decreases (refer to equation 1). This induces an EMF as per equation 2. DC motors function according to the same principles, further explored later in this section.

Equation 2 can be generalized for a coil with a wound number of turns N if the turns are identical and flux varies at the same rate through each turn:

$$\varepsilon = -N \frac{d\Phi_B}{dt} \quad (\text{eq. 3})$$

Lorentz Law

DC motors operate on the physical principle of Lorentz force, where a current-carrying conductor placed in a magnetic field experiences a force:

$$F = ILB \sin \theta \quad (\text{eq. 4})$$

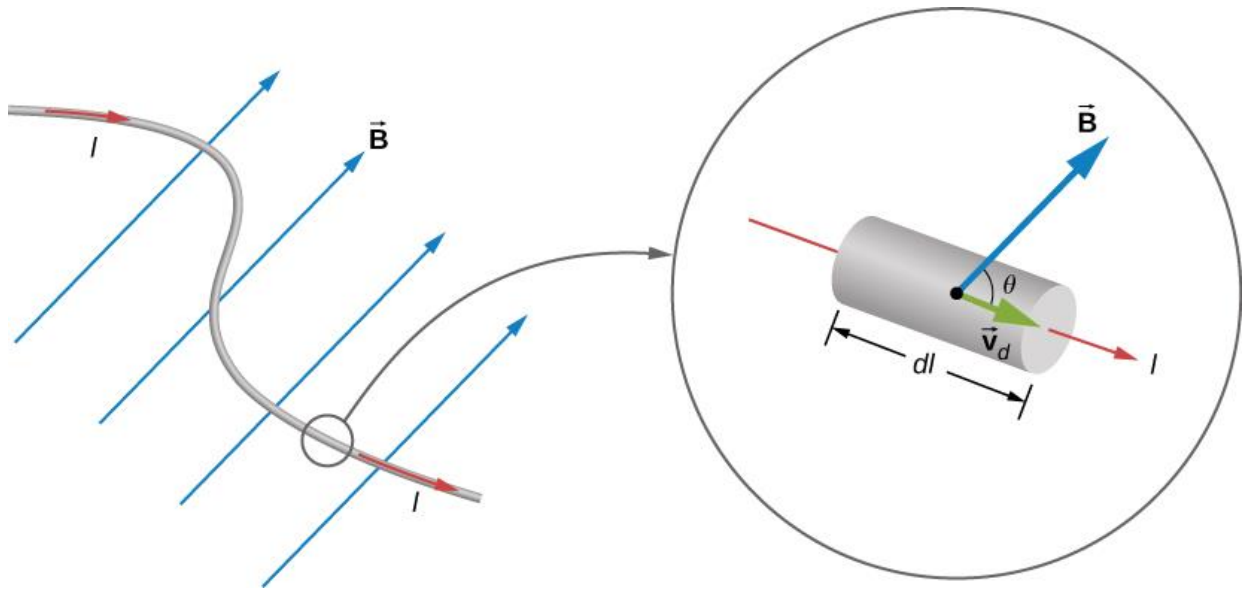


Figure 1.2: Force on a current-carrying wire in a uniform magnetic field (Physics Libretexts, 2025).

Where,

- I = current flowing through the wire (A)
- L = length of the wire (m)
- B = magnetic field strength (T)
- θ = angle between the current direction and magnetic field lines

The magnetic force on the wire segment is proportional to the component of B that is perpendicular to it. The direction of this force can be determined using Fleming's left-hand rule.

Components of a DC Motor

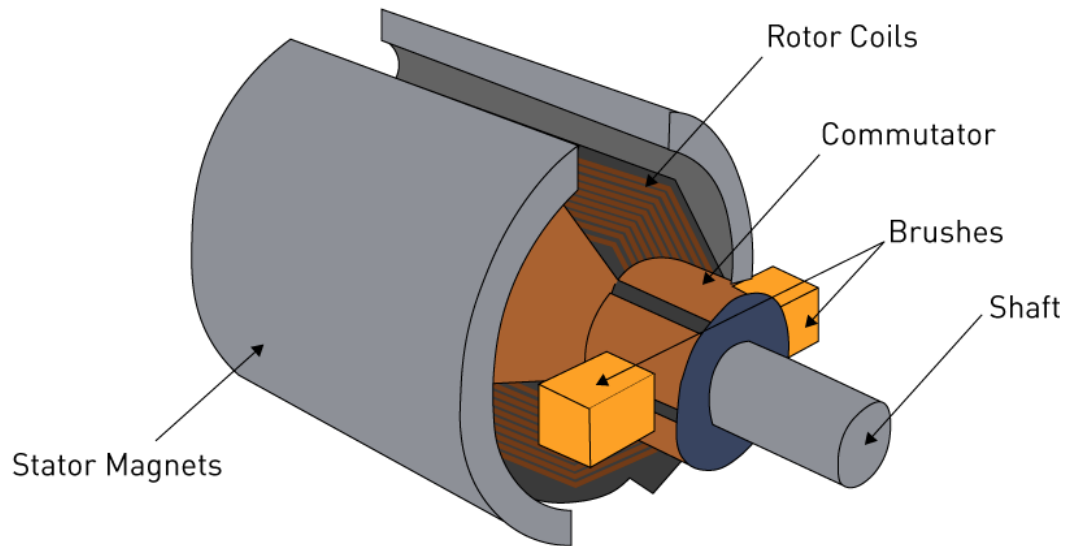


Figure 1.3: A diagram of the main components of a brushed DC motor (Monolithic Power Systems, n.d.)

The main components of a brushed DC motor are the stator, rotor, brushes, and shaft. The moving part of the motor is the rotor or armature that spins about the axis which is the shaft and comprises of tightly wound coils through which current from a power supply runs. The ends of the rotor are attached to circular conducting segments that form a commutator. In Figure 1.3 above, each commutator segment contacts one of the brushes of an external circuit that includes a power source (Young et al., 2020, p. 902).

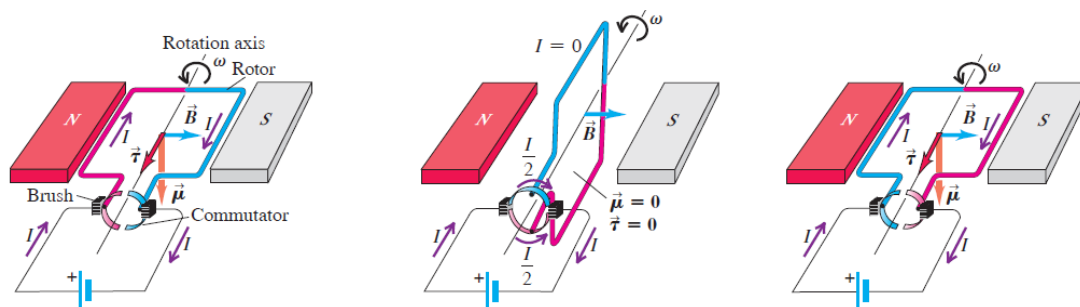


Figure 1.4: A diagram showing the functionality of a DC motor (Young et al., 2020, p. 902)

When direct current flows through the rotor, a magnetic moment $\vec{\mu}$ is generated as shown in Figure 1.4 on pg. 8. The rotor lies between two opposing ends of magnets, with a magnetic field $\vec{B}$ that exerts a torque $\vec{\tau} = \vec{\mu} \times \vec{B}$ on the rotor. The current switches direction every 180° when the brushes contact the other segment of the commutator, ensuring that the torque is always in the direction to rotate the rotor in the same direction. The motor reaches “steady-state” speed when the average magnetic torque is balanced by an opposing torque from friction in the rotor bearings, between the commutator and brushes, and due to air resistance.

Simulating a DC Motor

Modelling the dynamic behaviour of a DC motor involves describing its electrical and mechanical characteristics mathematically.

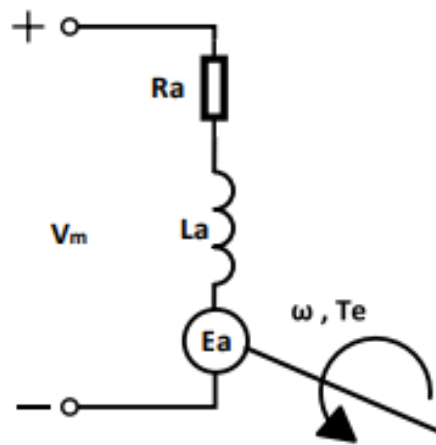


Figure 1.5: An equivalent circuit model of a DC motor

V_m	= Motor terminal voltage (V)
L_a	= Armature winding inductance (H)
R_a	= Armature winding resistance (Ω)
i_a	= Armature Current
ε	= Motor back EMF voltage (V)
K_b	= EMF constant ($\text{V} \cdot \text{s} \cdot \text{rad}^{-1}$)
K_t	= Torque constant ($\text{Nm} \cdot \text{A}^{-1}$)
τ_e	= Torque generated by motor ($\text{N} \cdot \text{m}$)
τ_L	= Torque load ($\text{N} \cdot \text{m}$)
τ_f	= Motor frictional torque ($\text{N} \cdot \text{m}$)
ω	= rotational speed ($\text{rad} \cdot \text{s}^{-1}$)
b	= dampening coefficient ($\text{Nm} \cdot \text{s} \cdot \text{rad}^{-1}$)
J	= Moment of inertia of rotor ($\text{kg} \cdot \text{m}^2 \text{s}^{-2}$)
θ	= Rotor position (rad)

The electrical equation of the DC motor in Figure 1.5 above according to Kirchhoff's voltage law is:

$$V_m = R_a i_a + L_a \frac{di_a}{dt} + \varepsilon \quad (\text{eq. 5}) \quad (\text{Altay and Yildiz, 2022, p. 183})$$

The induced back EMF is:

$$\varepsilon = K_b \omega \quad (\text{eq. 6})$$

Substitute equation 6 into equation 5:

$$V_m = R_a i_a + L_a \frac{di_a}{dt} + K_b \omega \quad (\text{eq. 7})$$

This gives the electrical component for modelling the simulation of the DC motor. Modelling the mechanical component involves determining the forces driving and opposing the motor's rotation. Newton's second law in rotational motion gives:

$$\tau_M - \tau_L - \tau_f = J \frac{d\omega}{dt} \quad (\text{eq. 8})$$

The torque generated by the motor is given by:

$$\tau_M = K_t i_a \quad (\text{eq. 9})$$

The frictional constant is given by:

$$\tau_f = b\omega \quad (\text{eq. 10})$$

The load torque is given by:

$$\tau_L = r F_{load} \sin \phi \quad (\text{eq. 11})$$

Where r is the radius of the shaft, F_{load} is the force the load exerts on the lever arm, and ϕ is the angle between $\vec{r}$ and $\vec{F}$. Substituting equations 9, 10, and 11 into equation 8 yields:

$$K_t i_a - r F_{load} \sin \phi - b\omega = J \frac{d\omega}{dt} \quad (\text{eq. 12})$$

Equation 12 is used to simulate the mechanical component of a motor on MATLAB. Finally, the torque constants will be used to vary the magnetic field strength. The torque constant, K_t , is directly proportional to the magnetic field strength (Johnson, 2005, Chapter 8, p. 3) and equal in value to the back EMF constant K_b . This allows for altering the magnetic field strength in the MATLAB simulations by altering the back EMF and torque constants.

To generate a second simulation to incorporate thermal losses, an equation for the resistance changes from R_o to R as the temperature changes from T_o to T is required, which is as follows:

$$R = R_o[1 + \alpha(T - T_o)] \quad (\text{eq. 13}) \quad (\text{Physics LibreTexts, 2025, Section 9.4})$$

Where α is the temperature coefficient of resistivity. Note that equation 13 is only accurate for small temperature changes. Iron losses are not considered due to their complexity.

Magnetic saturation occurs when the external magnetic field cannot increase the magnetization of the material further, so the total magnetization affecting the material levels off. To simulate this, A hyperbolic tangent relationship (Herceg et al., 2020, p. 2) in the form of:

$$K_{t\,ref} = \tanh\left(\frac{i_a}{I_{sat}}\right) \quad (\text{eq. 14})$$

where I_{sat} is the “knee” current where saturation starts, and $K_{t\,ref}$ is a multiplication factor to determine the magnetic field strength where saturation is considered, given by $B_{eff} = B \cdot K_{t\,ref}$.

This means that increases in the magnetic field strength B no longer result in proportional increases in the actual magnetization affecting the rotor, so the growth of the torque and back EMF constants decreases.

3-Experimental Variables

Independent Variable:

The magnetic field strength (B/T) between the rotor and the magnet ranged from 0.030T to 0.100T and was changed by varying the distance of the air gap between the rotor and the magnets, as shown in Figure 2.6 on pg. 18.

Dependent Variable:

The mechanical power (P/W) generated by the motor, which was obtained by using a known mass to serve as the load to measure the torque produced by the motor and by measuring the rotational speed ω measured in rotations per minute (RPM), to calculate the power $P = \tau \cdot \omega$ (eq. 15) (Young et al., 2020, p. 315).

Controlled Variables:

This section discusses the controlled variables of this experiment, outlining how and why they were controlled.

The type of magnets and supports were controlled by using identical neodymium magnets and metal vices. Varying magnet shapes, sizes, and types create varying magnetic field distribution areas, influencing magnetic flux according to equation 2.

Number of turns of copper wire on armature were controlled by fixing the number of turns of copper wire to be three hundred on each side. The number of turns N affects the induced EMF on the rotor according to equation 3.

The power input had to be controlled in conjunction with the temperature of the DC motor by keeping the power supply input at 40W and using a fan to cool down the coils to less than 27

degrees Celsius after every trial, which was measured using a non-contact temperature sensor.

Varying power inputs affect the rotational speed of the motor, and higher temperatures can decrease current and increase power loss due to heat, introducing systematic error unless the rotor cools down between trials.

The load of the motor had to be kept constant, because an increase in the load torque would subsequently increase the rotational speed. To visualize this, refer to $\tau = K_t i_a$ from equation 9.

A higher torque draws out more armature current, and this effect can be modelled at steady-state speeds using equation 7 at steady-state speeds where: $\frac{di_a}{dt} = 0$

$$i_a = \frac{V_m - \varepsilon}{R_a} \quad (\text{eq. 16})$$

An increase in current must be accompanied by a decrease in the back EMF of the motor, decreasing the rotational speed according to equation 6 (Precision Microdrives, 2022).

4–Experimental Design

4.1–Apparatus

- 1 wood plank (0.47m × 0.14m × 0.03m)	- 2 paper clips
- 2 ball bearings (0.04m inner ring diameter)	- Power supply (max 10A and 30V)
- 1 metal bolt (0.15m)	- 2 alligator clips
- 2 locknuts	- Laptop with PASCO application
- 26-gauge enameled copper wire	- PASCO Magnetic Field Sensor
- Electrical tape	- Non-contact tachometer and reflective tape
- 1 PVC pipe (0.04m × 0.30m)	- Non-contact temperature sensor
- 2 copper strips (0.06m×0.005m)	- Fan
- Superglue	- Fishing line (8.0m)
- Soldering iron and soldering wire	- Fishing rod
- 8 neodymium magnets	- Measuring tape
- 2 metal vices	- 0.0200kg weight
- Tools (screws, metal washers, drill, hammer, wire cutter, screwdriver, wood saw bench, sandpaper)	- Gloves
- 1 clamp	

Figure 2.1: A table detailing the experimental apparatus

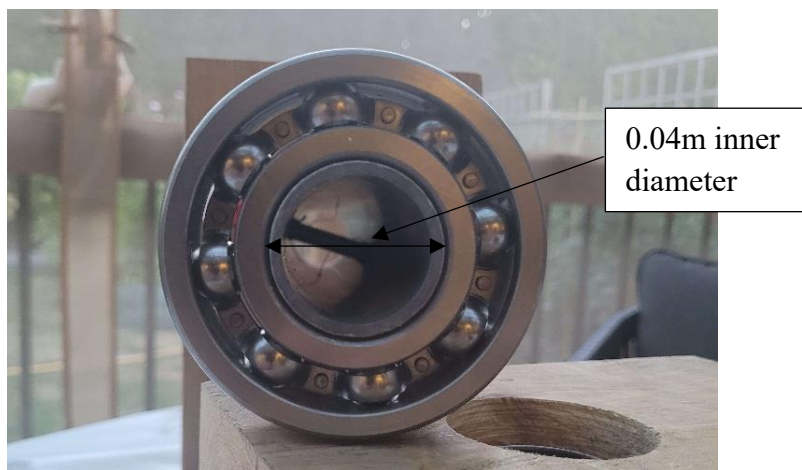


Figure 2.2: A photo of the ball bearing used for the motor

4.2–Method

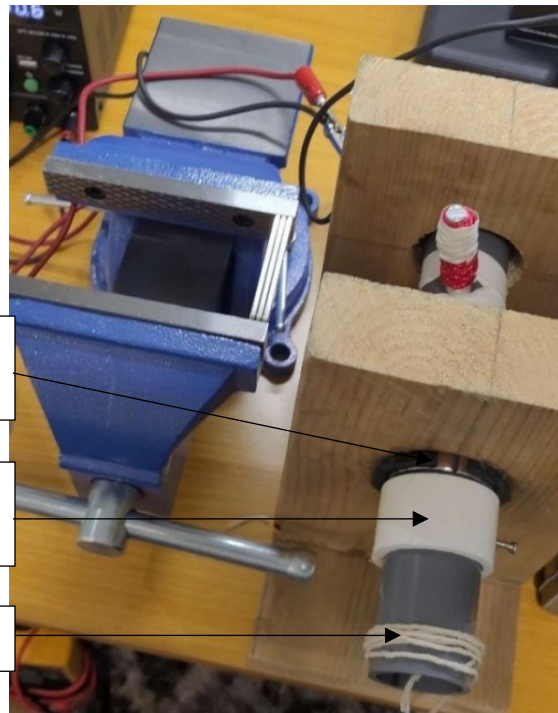


Figure 2.3: top-front-view of prototype motor



Figure 2.4: Photo of copper cylinder substitute for bearing in prototype motor

Preliminary Trials

A prototype motor was built to determine materials and parameters for main motor construction. 26-gauge copper wire was chosen for winding convenience. A copper cylinder with a clamping collar (see Figures 2.3 and 2.4 above) was originally used as a substitute for a bearing but was replaced with a metal ball bearing due to high frictional losses hindering motor performance. (other parameters discussed in Explanation for Apparatus section)

Experimental Setup

A wooden base ($0.28\text{m} \times 0.14\text{m}$) was prepared using a wood saw bench. Two additional wooden pieces ($0.20\text{m} \times 0.14\text{m}$) were used as vertical supports. One support was screwed perpendicularly to the base.

Two concentric holes with an outer diameter equal to the outer diameter of the ball bearings were drilled 0.05m down from the top edge of the supports. The ball bearings were then fitted onto the ends of a PVC pipe by using a hammer and then aligned with the holes on the attached support. The other support was subsequently positioned and fixed on the opposite end of the base so that the bearings were concentrically aligned.

A hole was drilled through the center of the shaft, then a 0.15m metal bolt was inserted and secured with metal nuts on both sides to form the rotor. Copper wire was tightly wound around the bolt 300 times on each side in the same direction, leaving the two ends of the wire loose. The enamel coating on these ends was removed using sandpaper, and the ends were soldered to two copper strips.

The copper strips were bent and superglued to the shaft near the support, ensuring a small gap between their ends and that the gap aligned with the rotor (Figure 2.8 on pg. 19). This served as the commutator. Two screws with metal washers were drilled into the wooden support above the commutator, and paper clips were placed beneath the washers and bent to maintain constant contact with the commutator surface, acting as brushes. The paper clips should be positioned on the direct opposite side of the commutator to each other.

A fishing line was wrapped twice around the shaft and secured using superglue. The line was threaded through the hole of a fishing rod, and a 0.0200kg mass was attached to its free end (see Figure 2.7 on pg. 18). The apparatus was positioned so that the suspended load hung at least 4m above the ground.

A DC power supply was connected to the motor by connecting two alligator clips to the two screws acting as brush terminals. Two metal vices, each holding four neodymium magnets,

were placed on either side of the rotor with opposing poles facing each other. Reflective tape was attached to the shaft away from the rotor and commutator. The DC motor was clamped securely to the table to prevent shifting during operation.

Data Collection

The magnetic field strength between the magnets and rotor was adjusted by repositioning the metal vices and using a PASCO magnetic field sensor to ensure a magnetic field strength of 0.030T on each side of the rotor, measured from its tip, (see Figure 2.6 on pg. 18).

The power supply voltage was set to its maximum of 30V, and the current limit was adjusted so that power input, as indicated on the power supply, was 40W. A non-contact tachometer was used to measure the rotor's steady-state speed, which was determined to be when fluctuations on tachometer RPM readings did not exceed 5 RPM. Five independent trials were performed to improve accuracy and reliability of the data. Mechanical power output was then calculated from rotational speed data using equation 15. Between trials, a fan was used to cool the coils to under 27°C, verified using a non-contact temperature sensor, and the fishing line was unwound to lower the load to ground level. The data collection procedure was repeated at magnetic field strengths ranging from 0.030T to 0.100T and increased in increments of 0.010T by reducing air gap between the magnets and rotor (see Figure 2.6 on pg. 18).



Figure 2.5 (above): A picture of the experimental setup and DC motor

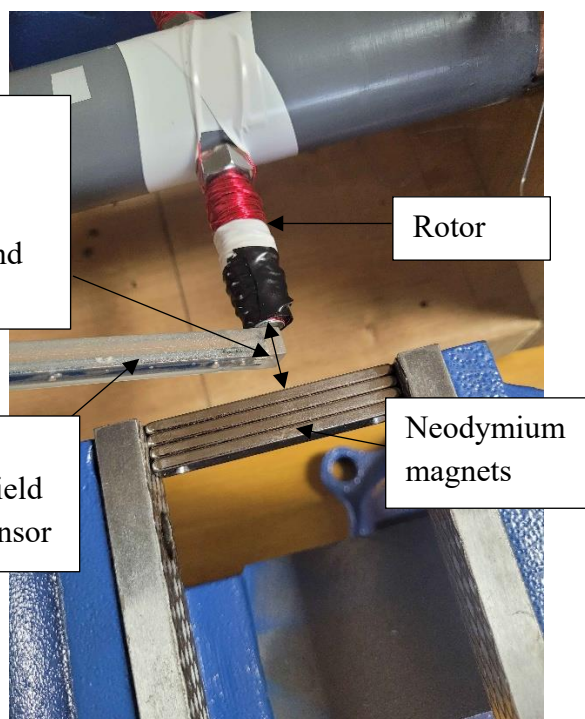


Figure 2.6: Photo showing air gap and reference position for magnetic field strength measurements

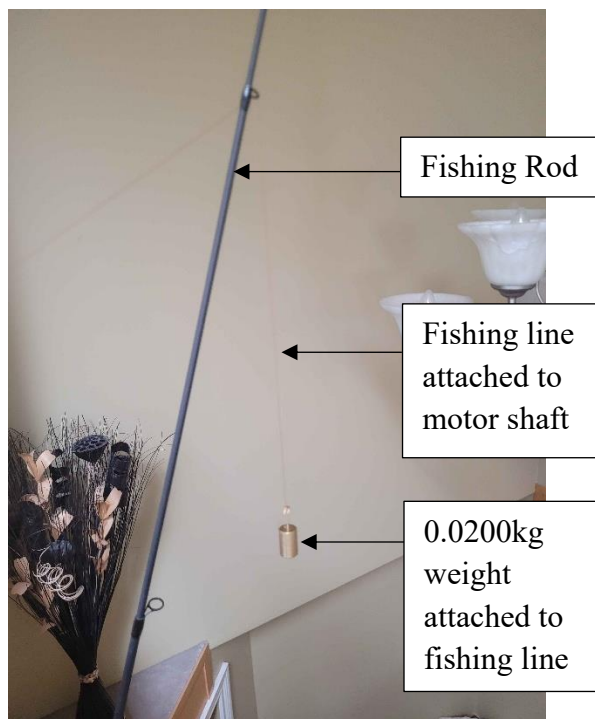


Figure 2.7: Photo showing fishing rod orientation for experimental setup

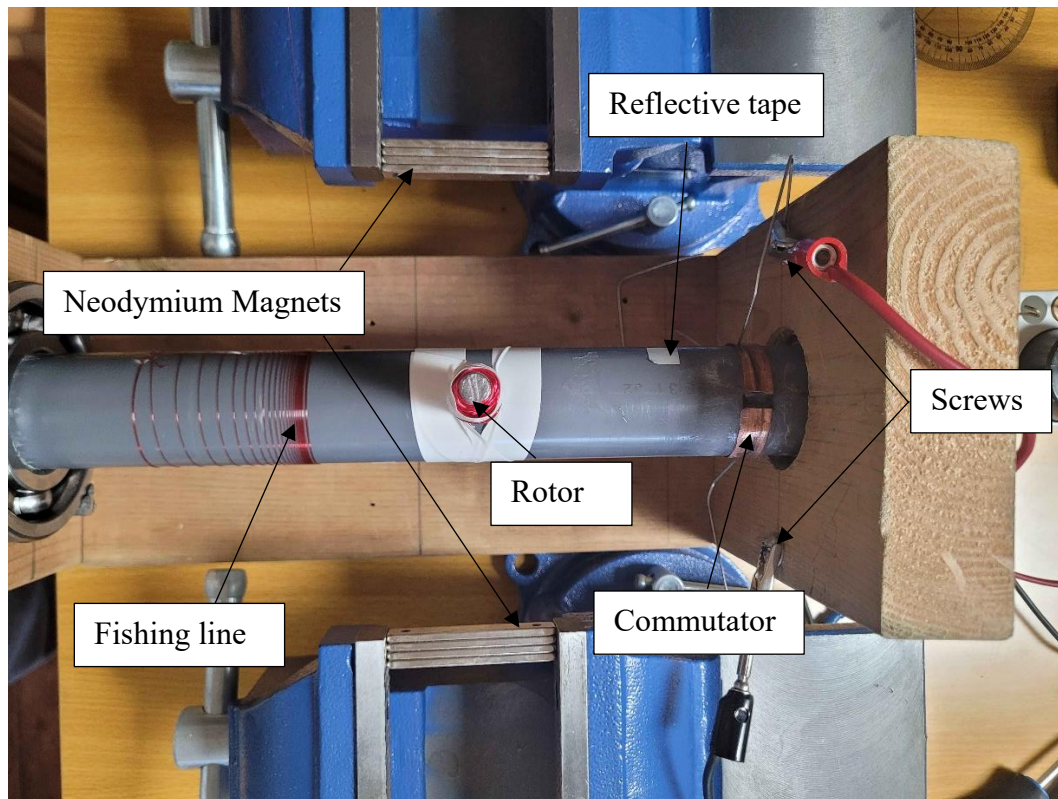


Figure 2.8: Photo of the top-view of the DC motor

4.3–Explanation for Apparatus

The power supply was set to 40W to allow the motor to run at lower magnetic field strengths whilst not supplying too much current to reduce heating effects. The magnetic field strength was set from 0.030T to 0.100T because at lower values, the motor would not run, and at higher values, the air gap was too narrow to measure the field strength. The use of ball bearings removes the need for a clamping collar, but it made construction more difficult since the shaft had to be sanded down and the bearing hammered on due to its perfect fit. The load on the motor was positioned at least 4m above the ground so the motor had time to reach steady-state speed, and its mass chosen to be 0.0200kg to avoid overloading which could cause excessive heating, and underloading which could render it difficult to measure the rotational speed before the load reached the shaft. Reflective tape was used for non-contact tachometer measurements.

4.4–Safety Information

Insulating gloves are necessary to prevent burns when using the soldering iron and when touching the brush gear of the motor, which can reach high temperatures of 50°C. A mask and protective eyewear should be worn when working with toxic soldering iron fumes. Caution should be taken when handling magnets so as not to crush one’s fingers. The experiment should be done on a dry, stable surface, since some of the equipment is heavy. Direct contact with electrical components should be avoided when power supply is running and rotor is rotating.

5–Raw Data Collection

Magnetic Field Strength (B/T) ($\pm 0.003T$)	Rotational Speed (ω/RPM) (± 5 RPM)					
	Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Trial 6
0.030	163	164	144	154	170	162
0.040	228	213	231	224	220	227
0.050	274	301	290	288	295	292
0.060	328	325	344	316	340	343
0.070	356	355	345	353	369	364
0.080	383	384	386	386	381	381
0.090	401	401	411	392	398	405
0.100	419	406	403	417	404	417

Figure 3.1: Raw data of rotational speed at different magnetic field strengths

Uncertainties for magnetic field strength and rotational speed from Figure 3.1 above are based on observed fluctuations in the instrument readings during experimentation.

6–Processed Data

6.1 - Experimental Data

Figure 4.1 below shows average rotational speed, power output, and their respective absolute uncertainties calculated from the raw data (see Figure 3.1 on pg. 20) in relation to the magnetic field strength.

Magnetic Field Strength (B/T) ($\pm 0.003T$)	Average Rotational Speed (ω_{avg}/RPM)	Absolute Uncertainty on Rotational Speed ($\Delta\omega/RPM$)	Motor Power Output (P/W)	Absolute Uncertainty on Motor Power Output ($\Delta P/W$)
0.030	160	13	0.066	0.007
0.040	224	9	0.092	0.006
0.050	290	14	0.119	0.009
0.060	333	14	0.137	0.010
0.070	357	12	0.147	0.009
0.080	384	3	0.157	0.006
0.090	401	10	0.165	0.009
0.100	411	8	0.169	0.008

Figure 4.1: Processed data of average rotational speed and power output at different field strengths

Sample Calculations

Sample calculations for the average rotational speed and mechanical power output at 0.030T:

$$\omega_{avg} = \frac{163 + 164 + 144 + 154 + 170 + 162}{6} = 159.5 \text{ RPM} \approx 160 \text{ RPM}$$

Random uncertainty was estimated using the half-range method:

$$\Delta\omega_{avg} = \frac{\omega_{max} - \omega_{min}}{2} = \frac{170 - 144}{2} = 13 \text{ RPM}$$

$$P = \tau \cdot \omega = rF_{load} \sin \phi \cdot \omega = r \cdot mg \sin \phi \cdot \omega$$

$$= 160 \text{ RPM} \cdot 2\pi \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot 0.0200 \text{ m} \cdot 0.0200 \text{ kg} \cdot 9.81 \text{ ms}^{-2} \cdot \sin 90^\circ = 0.0657 \text{ W} \approx 0.066 \text{ W}$$

Random error on rotational speed was propagated into power calculations to reflect random variations in measured values:

$$\Delta P = \left(\frac{\Delta r}{r} + \frac{\Delta m}{m} + \frac{\Delta \omega}{\omega} \right) \cdot P$$

$$\Delta P = \left(\frac{0.0005\text{m}}{0.0200\text{m}} + \frac{0.0001\text{kg}}{0.0200\text{kg}} + \frac{13\text{RPM}}{160\text{RPM}} \right) (0.0657\text{W}) = 0.006077\text{W} \approx 0.006\text{W}$$

6.2 – Theoretical Data

The theoretical data was obtained from simulations using MATLAB Simulink, which models the behaviour of a DC motor using equations 5 to 13. The rotational speed was found by solving equations 8 and 12 at different magnetic field strength inputs. The theoretical motor model with thermal losses was generated using equation 13 and the saturation model was based on equation 14, allowing for more realistic comparison with theoretical data. All parameters for the simulation model were based on actual properties specific to the experimental motor. (refer to appendices A1, A2, A3, A4, A5, A6, and A7).

Magnetic Field Strength (B/T)	Motor Power Output without Thermal Losses (P/W)	Motor Power Output with Thermal Losses (P/W)
0.030	0.109	0.093
0.040	0.144	0.129
0.050	0.176	0.159
0.060	0.205	0.187
0.070	0.231	0.213
0.080	0.254	0.237
0.090	0.275	0.259
0.100	0.295	0.277

Figure 4.2 (above): theoretical data for models with and without thermal losses (base and thermal models)

Magnetic Field Strength (B/T)	Motor Power Output with Magnetic Saturation (P/W)
0.030	0.085
0.040	0.116
0.050	0.142
0.060	0.165
0.070	0.185
0.080	0.203
0.090	0.219
0.100	0.233

Figure 4.3: Theoretical data for base model with magnetic saturation (saturation model)

7 - Analysis

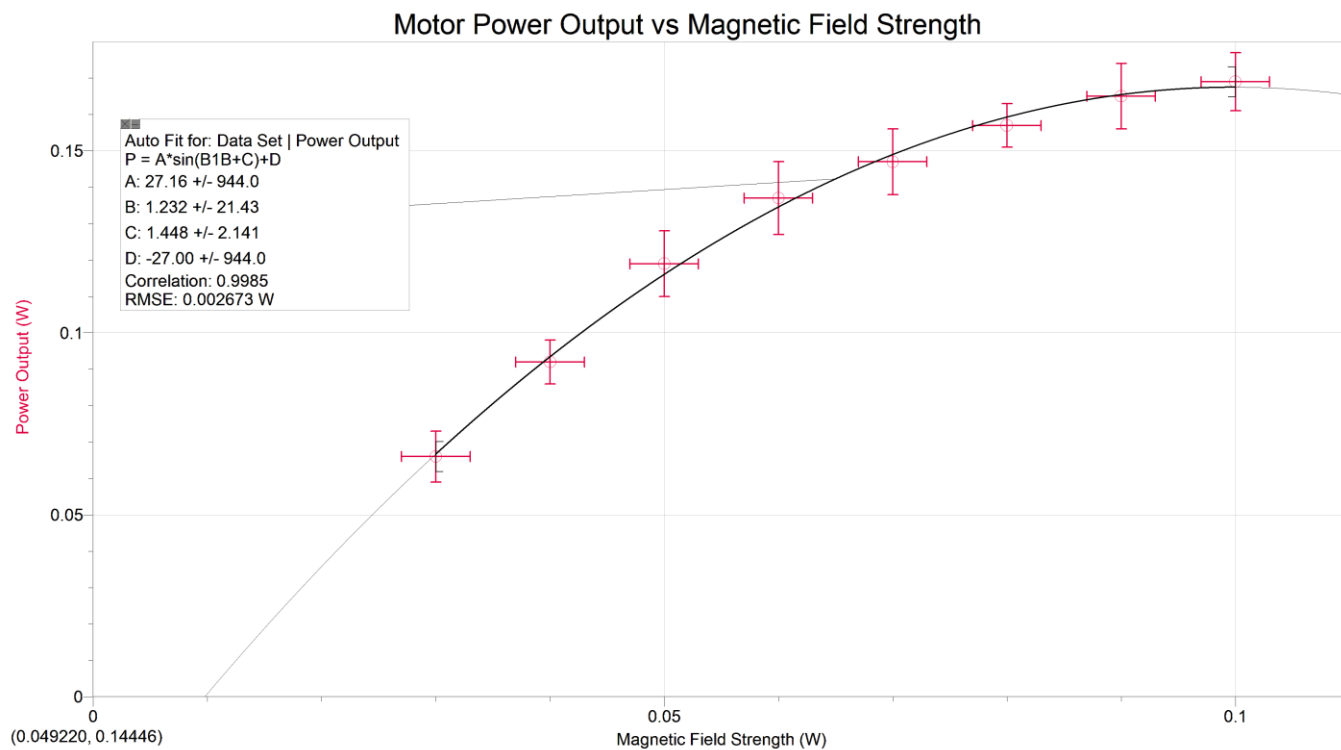


Figure 5.1: A graph of experimental power outputs against magnetic field strength

7.1 – Explaining the Trends

Logger Pro 3.16.2 with the built-in curve fitting tool was used, and a sinusoidal model was selected due to its high correlation (0.9985) and suitability for modelling the plateauing power output values at higher magnetic field strengths, further explained on the following page. Note that this equation is applied only to compare the shapes of curves and does not describe a relationship between magnetic field strength and motor power output, which follows complex interactions between current, magnetic flux, and torque. The curve in Figure 5.1 on pg. 23 is encompassed by the error bars of the data points, indicating that random error was properly accounted for.

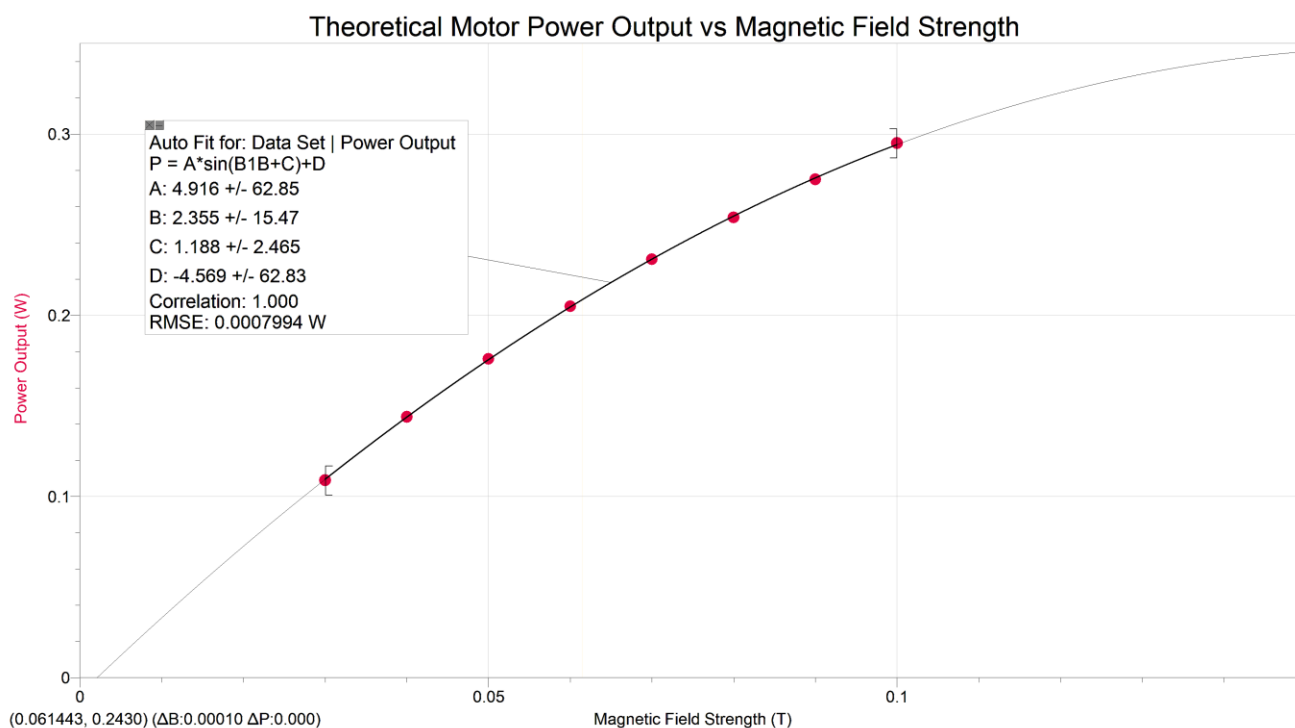


Figure 5.2 (above): A graph of the theoretical model's power output against magnetic field strength

The general trend of the data is that power output increases with increasing magnetic field strength, although the power output values plateau at higher magnetic field strength values. Additionally, the power output in the theoretical model in Figure 5.2 above increases at a faster

rate, reaching a maximum of 0.293W, whilst the experimental model in Figure 5.1 on pg. 23 only reaches 0.168W. This is due to thermal and resistive losses, and magnetic saturation present in the experimental model which is not considered by the base theoretical model.

The Increasing Trend

Increasing the magnetic field strength results in an increase in both the torque constant K_t and the back EMF constant K_b as discussed in the Background Information section. So as K_t increases, the torque increases according to $\tau_m = K_t i_a$ from equation 9. This allows the motor to overcome other frictional and opposing torques, increasing rotational speed and power output.

The back EMF constant K_b increases, causing the back EMF to increase according to $\varepsilon = K_b \omega$ from equation 6. Also, at steady-state speed, the electrical component follows equation 16 where the armature current is $i_a = \frac{V_m - \varepsilon}{R_a}$ or:

$$i_a = \frac{V_m - K_b \omega}{R_a} \quad (\text{eq. 17})$$

The numerator $V_m - K_b \omega$ decreases in value as K_b increases, so the armature current decreases.

The decreased current decreases the motor torque as per $\tau_m = K_t i_a$. At higher magnetic field strengths, greater rotational speeds increase the back EMF according to $\varepsilon = K_b \omega$, further decreasing current. Consequently, the rate of increase of power output diminishes at higher fields strengths and speeds, as seen in Figures 5.1 and 5.2 on pages 23 and 24 respectively. This phenomenon of back EMF limiting current and torque can be described as EMF dominance.

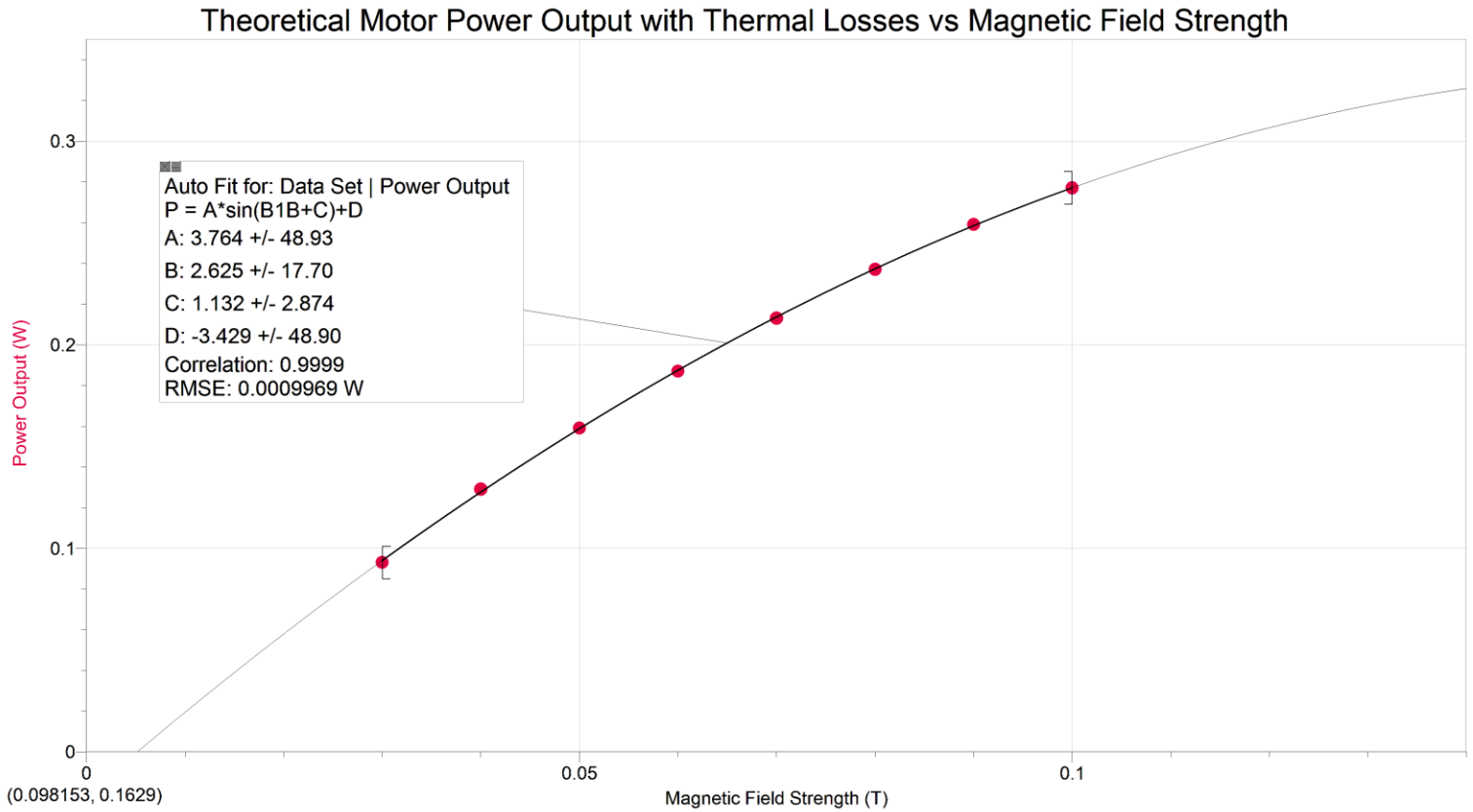


Figure 5.3: A graph of the theoretical (with thermal losses) power output against magnetic field strength

The theoretical model with thermal losses shown in Figure 5.3 above was used to help explain the effects of thermal losses on DC motor power outputs. Copper or I^2R losses cause heating in the copper wire of the DC motor. Resistance increases with temperature according to equation 13, causing the armature current i_a to decrease, resulting in decreased torque, rotational speed, and consequently power outputs. Additionally, not all thermal losses were considered, as iron losses such as hysteresis losses, Eddy current losses, and anomalous losses (Hargreaves et al., 2013, p. 1) were ignored due to their complexity, likely leading to a significant overestimate of power output values in the thermal model.

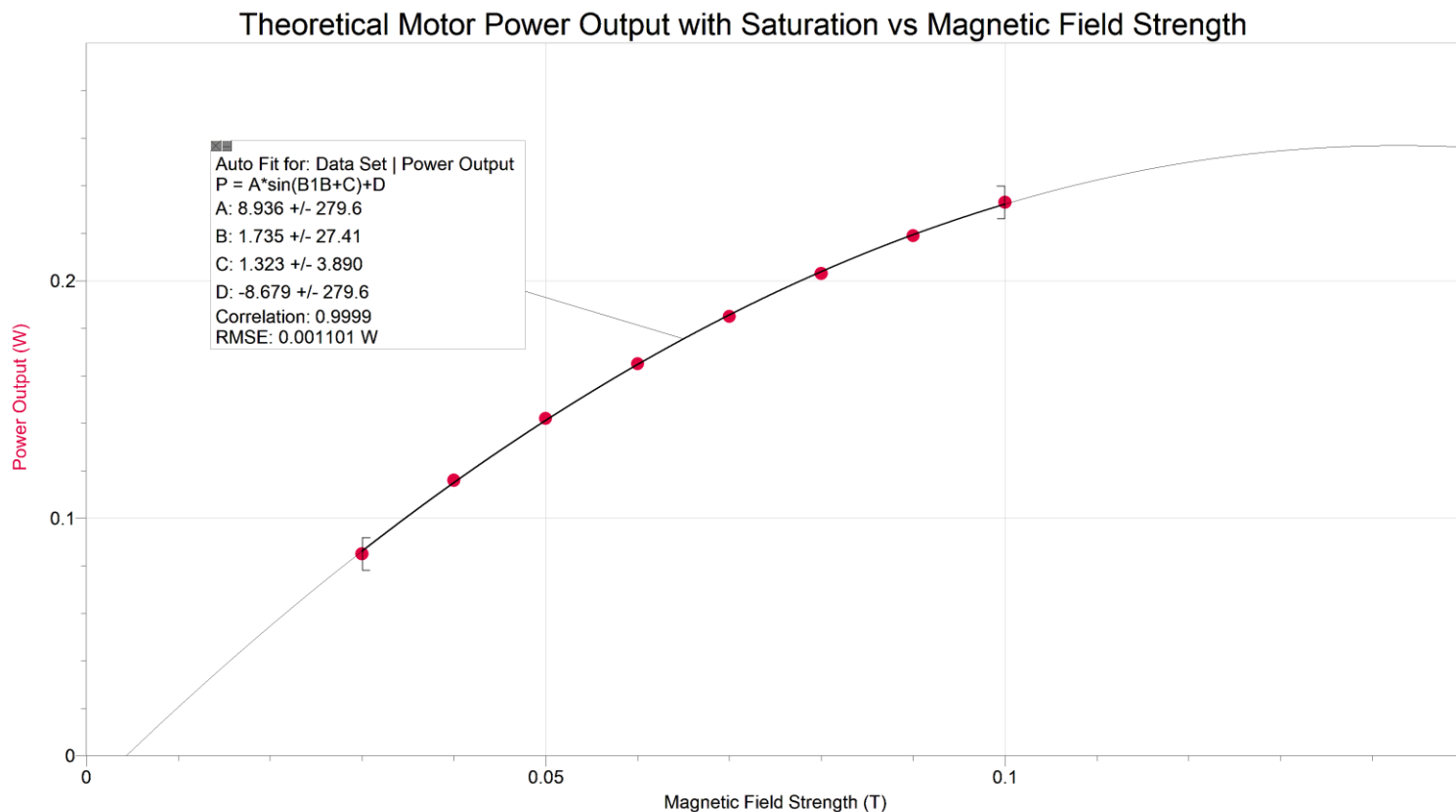


Figure 5.4: a graph of the theoretical (with saturation) power output against magnetic field strength

Figure 5.4 above shows data obtained from the base theoretical model (see Figure 5.2 on pg. 24) but with magnetic saturation being considered. Compared to the base theoretical model, the power output values level out faster, only reaching 0.233W compared to the 0.293W reached by the base theoretical model.

7.2 - Comparison of Experimental Data and Theoretical Models

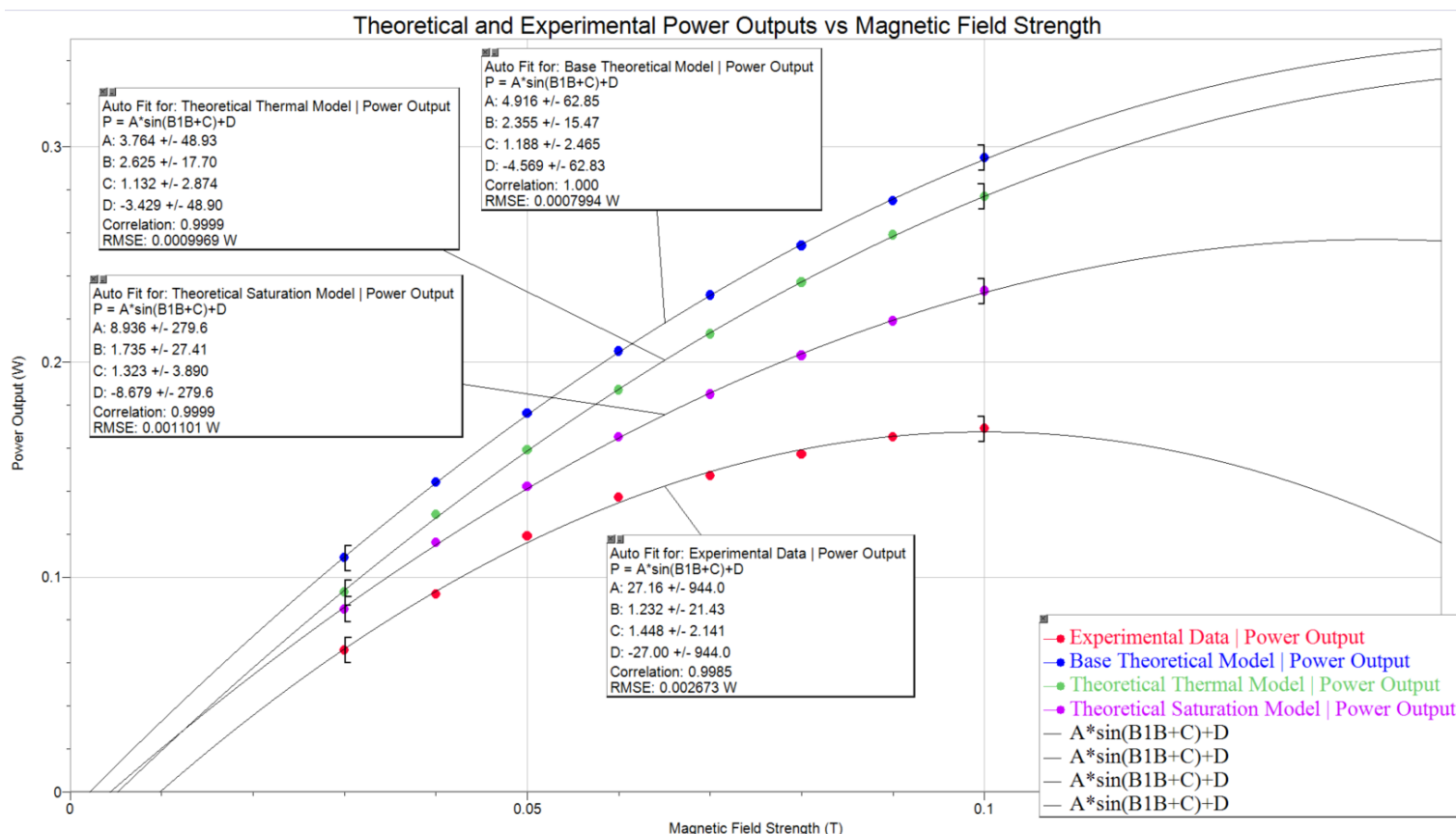


Figure 5.5: A graph of all the theoretical and experimental power outputs against magnetic field strength

In Figure 5.5 above, it can be noticed that the range of power output values for all the theoretical models is greater, since the experimental model suffers from additional resistive losses in the commutator, brush gear, and due to iron losses (Eddy current and hysteresis losses).

Additionally, the shapes of the different curves can be compared using constants from the sinusoidal equation used to model these curves, given by $P = A \sin(B_1 \cdot B + C) + D$ where P is the power output, B is the magnetic field strength, and A , B_1 , C , and D are constants. Comparing the different values of the constant B_1 , which scales the graphs horizontally, differences are evident between the theoretical and experimental data. The base and thermal theoretical models have relatively larger B_1 values of 2.355 and 2.625 respectively, whilst the theoretical saturation

model has a lower value of 1.735. Smaller values of B_1 define a more compressed curve, meaning power output values level off at lower magnetic field strength values.

The experimental data, with a B_1 value of 1.232, is the closest in shape to the curve of the theoretical saturation model, indicating that magnetic saturation effects better describe the experimental motor's behaviour than the base or thermal theoretical models. Remaining discrepancies between theoretical models and experimental data may have resulted from the additional resistive and iron losses mentioned previously, which are not considered in the theoretical models.

Although all the curves begin to plateau at higher magnetic field strengths, a definitive maximum power output value cannot be obtained due to the lack of range of magnetic field strength values, since the range of B values able to be tested was limited in this experiment.

8 – Evaluation and Conclusion

This experiment aimed to determine how varying the magnetic field strength of a DC motor affected its mechanical power output. It was successfully shown that an increase in magnetic field strength leads to an increase in the mechanical power output, although this increase is limited at high magnetic field strengths due to magnetic saturation, and resistive & thermal losses. These results highlight the importance of optimizing magnetic conditions in DC motor applications, although a definitive conclusion for optimal field conditions could not be obtained due to limitations encountered during experimentation.

Observations in the theoretical models matched that of the experimental models, although slight inaccuracies were evident, such as deviations in the shapes of the curves shown in Figure

5.5 on pg. 28, and the consistently lower maximum power output values of the experimental data which reached $0.169 \pm 0.008 \text{ W}$, compared with the 0.293 W reached by the base theoretical model for example. These discrepancies were explained in the Analysis section (pg. 23 to 29)

The main limitation in this experiment was the range of magnetic field strength values that could be tested. Reducing the air gap between the magnets and the rotor further would render measurement impossible, since the gap is too small to fit the PASCO magnetic field sensor. Stronger magnets must be used to achieve greater magnetic field strength values without having to narrow the air gap as much. Due to this limitation, observed power output values were restricted so peaks in their values could not be observed on graphs, preventing a complete understanding of saturation effects. This also made it difficult to accurately determine the saturation “knee” I_{sat} in equation 16 to model magnetic saturation (see Figure 5.4 on pg. 27) which likely led to overestimating the motor’s theoretical maximum power output.

Issues in the construction of the motor were also evident, observed in the sparking commutator, causing fluctuations in current, rotational speed, and subsequently calculated power output values, significantly increasing random error, with percentage uncertainties reaching 11% (refer to Appendix A8), though all other values stayed below 10%, indicating reasonably high precision. However, random error on power outputs could’ve been easily reduced had a vernier caliper been used to measure motor shaft diameter. Measuring the magnetic field strength itself also proved difficult, since even slight movements of the sensor caused significant fluctuations in the readings, contributing to random error as high as 10% at low field strengths (refer to Appendix A8). The load torque also swung as it was pulled up, generating inconsistent load torques and currents, though this source of random error was observed to be less significant than others.

Furthermore, iron losses were ignored in the thermal model, whose absence likely led to underestimating impacts of thermal conditions on motor performance. Future studies could aim to include these losses in theoretical motor simulations; however, it is important to note that extra parameters must be measured, and calculations are higher in complexity.

Other future studies could analyze effects of changing other motor construction parameters such as the number of coils, the inertia of the shaft and armature, and the load on the motor to investigate optimal constructions of DC motors for real-world applications. Different types of DC motors and synchronous motors can be investigated to compare properties and determine their efficiency under different conditions such as varying load torques and power inputs.

DC motors are central to modern technology, being in appliances, medical equipment, and the electric vehicle industry. Understanding the effects of altering certain variables such as those explored in this experiment is crucial to improving DC motor efficiency and performance. Through advancements, engineers can design more efficient motors that are better suited to the demands of a sustainable, electrified future.

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10 – Appendices

10.1 - Appendix A1: Base Theoretical Simulink Model of DC Motor

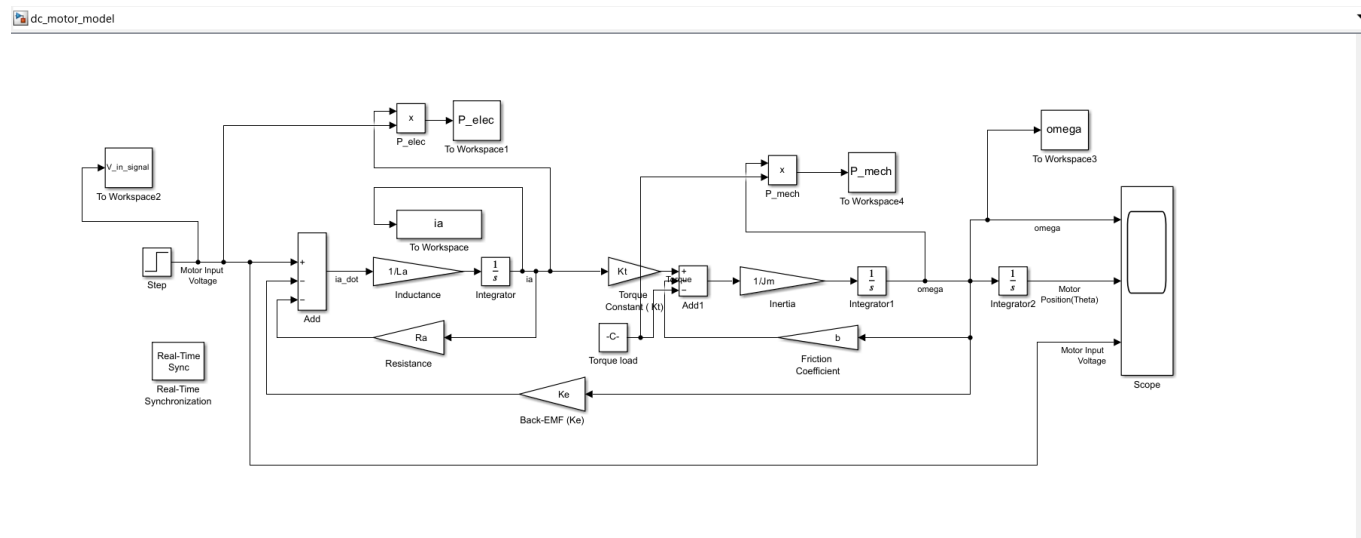


Figure 6.1 (above): A photo of the base theoretical DC motor Simulink model

10.2 - Appendix A2: Thermal Theoretical Simulink Model of DC Motor

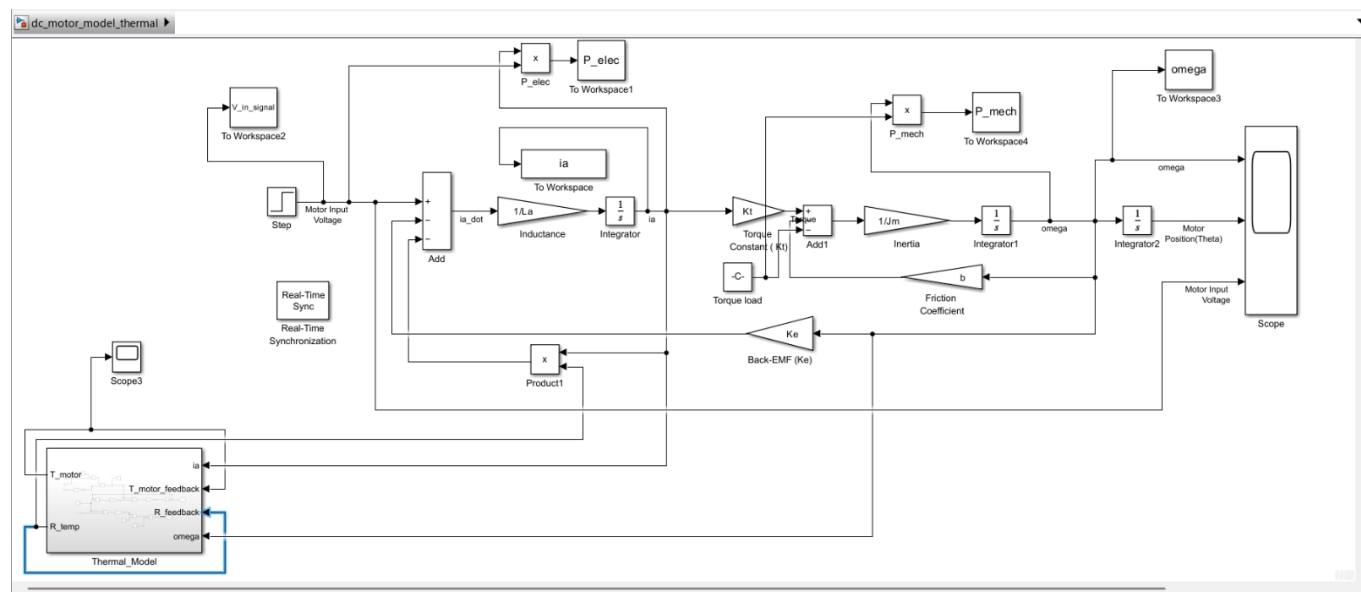


Figure 6.2: A photo of the thermal theoretical DC motor Simulink model

10.3 - Appendix A3: Thermal Subsystem of Thermal Theoretical Simulink Model of DC Motor

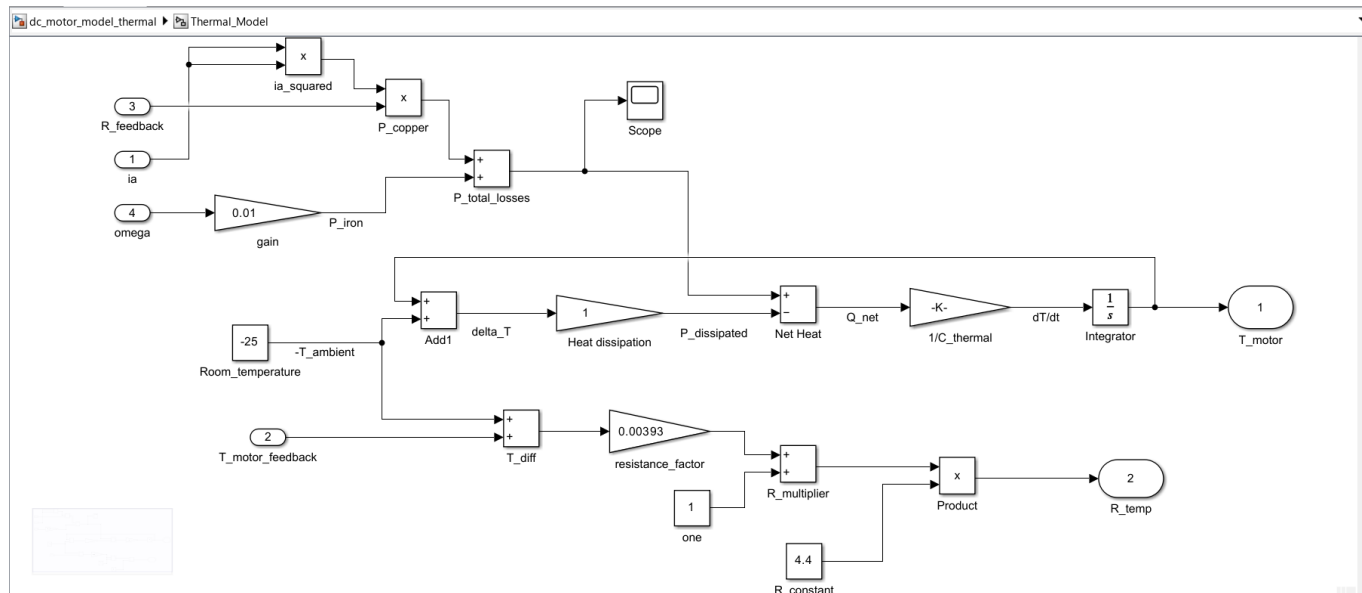


Figure 6.3 (above): A photo of the thermal subsystem of the thermal theoretical DC motor Simulink model

10.4 - Appendix A4: Base Theoretical DC Motor Parameters

```
>> %basic model parameters
Ra = 4.4;           % Motor Armature Resistance (Ohms)
La = 0.05;         % Motor Armature Inductance (H)
Jm = 0.00872;      % Moment of Inertia of Motor Rotor (kg.m2)
b = 0.003;         % Viscous friction coefficient
T_load = 0.00392;  %load torque (Nm)
B = 0.03;          %magnetic field strength (T)
B_ref = 1;
Kt_ref = 1;

Kt = Kt_ref * (B / B_ref); %torque constant
Ke = Kt;           %back emf constant

assignin('base', 'T_load', T_load);
assignin('base', 'Kt', Kt);
assignin('base', 'Ke', Ke);
assignin('base', 'P_elec', 40);
```

Figure 6.4: A photo of the C code for the base theoretical motors' parameters

10.5 - Appendix A5: Thermal Theoretical DC Motor Parameters

```
%thermal model parameters
Ra = 4.4; La = 0.05; Jm = 0.00872; b = 0.003; T_load = 0.00392;
B = 0.03; B_ref = 1; Kt_ref = 1; Kt = Kt_ref * (B / B_ref); Ke = Kt;
C_thermal = 6.9; temperature_coefficient = 0.00393;

% Assign variables to workspace
assignin('base', 'Ra', Ra);
assignin('base', 'La', La);
assignin('base', 'Jm', Jm);
assignin('base', 'b', b);
assignin('base', 'T_load', T_load);
assignin('base', 'Kt', Kt);
assignin('base', 'Ke', Ke);
assignin('base', 'P_elec', 4);
assignin('base', 'C_thermal', C_thermal);
assignin('base', 'temperature_coefficient', temperature_coefficient);

%run thermal model simulation
sim('dc_motor_model_thermal', 'StopTime', '45');
```

Figure 6.5: A photo of the C code for the thermal theoretical motors' parameters

Explanation for Parameters

- $R_a = 4.4 \, \Omega$ (ohms)

The resistance of the copper wires without the heating effect was measured using a multi-meter.

- $L_a = 0.05\text{H}$ (Henry)

The inductance is calculated according to the equation $L = \frac{N^2 \mu A}{l}$ to be 0.051H where N is the number of coil turns, μ is the magnetic permeability of the core material, A is the inner core area in m^2 , and l is the length of the coil in m.

- $J_m = 0.00872\text{kg/m}^2$

The moment of inertia of the motor was calculated according to the equation $\frac{1}{3}ML^2$. However, this does not consider the rotor protrudes from the shaft, and the inner part of the ball bearing. However, since this only affects the dynamical part of the DC motor (its angular acceleration), its effect is minimal.

- $b = 0.003$

The damping coefficient of the motor, which was estimated through research and trial and error during the simulation process. This value attempts to account for friction in the bearings, as well as friction due to air resistance.

- $B = 0.030\text{T}$ (Teslas)

The magnetic field strength of the motor, which ranged between 0.030T and 0.100T and was simulated by changing the values of K_t and K_e , which are the torque constants and back EMF constants respectively.

- $C_{\text{thermal}} = 6.9\text{J}/^\circ\text{C}$

This was calculated by multiplying the specific heat capacity of copper with the mass of the copper wire in the windings: $C_{\text{thermal}} = 385 \cdot 0.018 = 6.93 \approx 6.9\text{J}/^\circ\text{C}$

- $\text{temperature_coefficient} = 0.00393$

The temperature coefficient of copper which represents that for every 1°C increase in temperature, the electrical resistance of copper increases by 0.393%.

- $\text{Room_temperature} = 25^\circ\text{C}$

The room temperature which was manually input in Simulink as shown in Figure 6.3 on pg. 35

10.6 - Appendix A6: Sample Calculations for Magnetic Field Strength Under Saturated Conditions

The following are the sample calculations for the magnetic field strength under saturated conditions for

$B = 0.030\text{T}$. Equation 14 yields:

$$K_{t\text{ref}} = \tanh\left(\frac{i_a}{I_{sat}}\right) = \tanh\left(\frac{2.986\text{A}}{2.6\text{A}}\right) = 0.8172$$

$$B_{sat} = K_{t\text{ref}} \cdot B = 0.8172 \cdot 0.030\text{T} = 0.024156\text{T}$$

The final value for B_{sat} was used for the magnetic field strength in the MATLAB Simulink DC motor simulation that is depicted in Figure 6.1 on pg. 34. This allows for the calculation of the theoretical DC motor's mechanical power output what considers the effects of magnetic saturation.

10.7 - Appendix A7: Rotational Speed Data from Theoretical Models

Magnetic Field Strength (B/T)	Rotational Speed of Base Model (ω_{base}/RPM)	Rotational Speed of Thermal Model ($\omega_{thermal}/\text{RPM}$)	Rotational Speed of Saturation Model (ω_{sat}/RPM)
0.030	266.435	217.714	237.491
0.040	350.688	283.633	313.303
0.050	428.639	345.938	387.081
0.060	499.380	401.710	455.998
0.070	562.960	451.252	519.720
0.080	619.893	494.797	578.191
0.090	669.730	533.080	630.683
0.100	714.450	566.522	678.506

Figure 6.6: A table showing the rotational speeds from the three theoretical models in this experiment.

10.8 – Appendix A8: Conversion of Absolute to Percentage Uncertainties

Magnetic Field Strength (B/T)	Percentage Uncertainty on Magnetic Field Strength ($\Delta B_{\%}/T$)	Average Rotational Speed (ω_{avg}/RPM)	Percentage Uncertainty on Rotational Speed ($\Delta\omega_{\%}/RPM$)	Motor Power Output (P/W)	Percentage Uncertainty on Motor Power Output ($\Delta P_{\%}/W$)
0.030	10%	160	8%	0.066	11%
0.040	8%	224	4%	0.092	7%
0.050	6%	290	5%	0.119	8%
0.060	5%	333	4%	0.137	7%
0.070	4%	357	3%	0.147	6%
0.080	4%	384	1%	0.157	4%
0.090	3%	401	2%	0.165	5%
0.100	3%	411	2%	0.169	5%

Figure 6:7: A table showing the percentage uncertainties for all absolute uncertainties from experimental data in Figure 4.1 on pg. 21

Sample Calculations

The sample calculations for the percentage uncertainties at 0.030T are as follows:

$$\Delta B_{\%} = \frac{\Delta B}{B} \cdot 100\% = \frac{0.003T}{0.030T} \cdot 100\% = 10\%$$

$$\Delta\omega_{\%} = \frac{\Delta\omega}{\omega} \cdot 100\% = \frac{13RPM}{160RPM} \cdot 100\% = 8.125\% = 8\%$$

$$\Delta P_{\%} = \frac{0.007W}{0.066W} \cdot 100\% = 10.6\% = 10\%$$